SAMPLE - NOT ISSUED. No Engagement Terms are incorporated and no commitment of any kind is made. Issued reports carry the governing terms version here.

Order identity

Field	Value
Client	Acme Labs
Order id	LQI-2026-Troesch-20-2026-08-29-T2
Reference	LQI-2026-Troesch-20
Problem	A plasma-confinement model.
Stated equation & data	y" = 20*sinh(20*y), y(0) =0, y(1) = 1
Ordered (date in)	2026-08-29
Delivered (date out)	2026-08-29
Service tier	Tier 2: Certified Standard

CERTIFICATE - Tier 2 (Certified Standard)

- Class: ODE boundary-value problem | Engine: SIRPY native — power series | Generated by SIRPY 0.59.15
- FINDING: solved, NOT verified
- max residual (measured): 4.851e+09 at x = 1
- artifact coverage: [0, 1]; pieces solved 64, delivered 64
- artifact: LQI-2026-Troesch-20-T2.artifact.json
- sha256: `7bfb7cf8 fffc1d8e 97d825d1 aa30b2a2 d30a3b2b 57f5531d b1cee451 6acebc25`
- reproduction: the headline numbers above are RECOMPUTABLE from this artifact and the stated equation using only Python and sha256sum -- substitute the delivered coefficients into the equation and compare. This is reproduction of our measurements, not agreement with a third-party solver: an independent solver may be compared against the delivered values, and any disagreement is a finding to report, not a covered reproducibility failure. Scope: the artifact coverage above.

Claim 1 — Constraint satisfaction (measured)

Equation residual by exact coefficient differentiation at dense off-node points: max 4.851e+09 (relative 9.999e-01).

Boundary/initial data residual: 7.749e-14.

Checked: the equation (residual by exact coefficient differentiation); boundary data.

Claim 2 — Seam continuity (measured)

Max one-sided jump across 63 interior node(s): 1.155e-06 - measured from the delivered coefficients, exactly at each node. The per-node figure in this package plots this same measurement, annotated with this same value.

Claim 3 — Measured tail decay and trust radius

y: coefficient tail/head ratio per piece - median 9.0e+01, worst 1.1e+20 over 64 piece(s); a MEASURED decay indicator, not a continuous-interval remainder bound.

See the footnote "How to read the tail decay ratio" at the end of this section.

Claim 4 — Independent forward check

exact energy-quadrature relation: 1.618e-03.

Characteristic values (from delivered data)

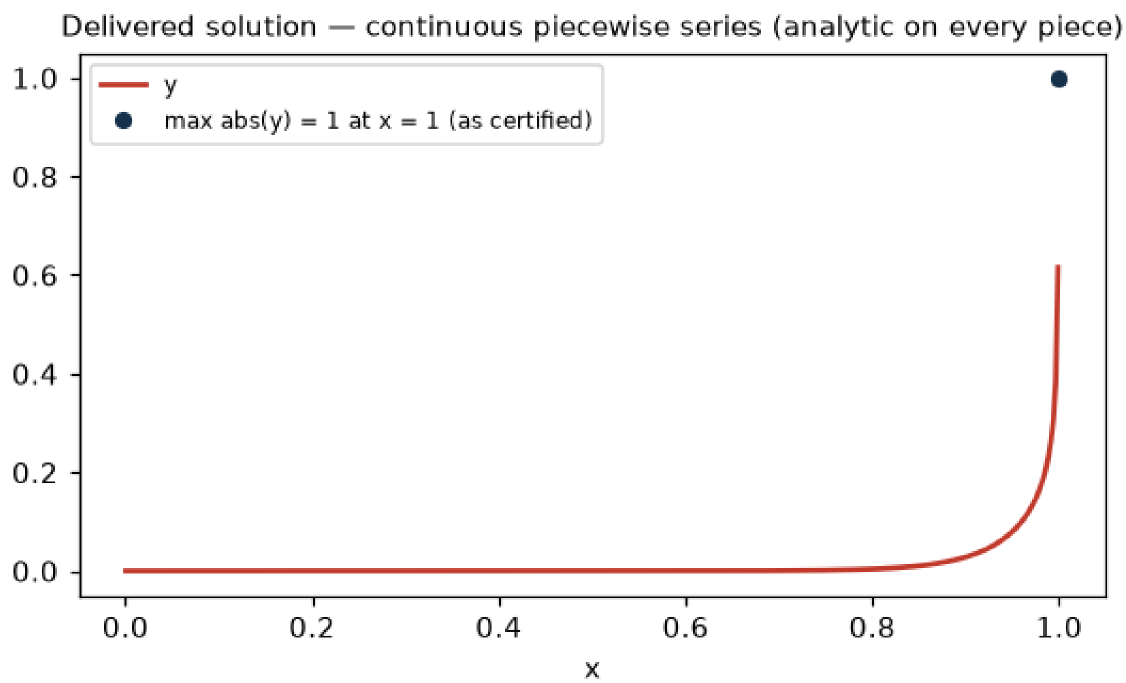
max abs(y) = 1 at x = 1 [computed from the delivered series representation]

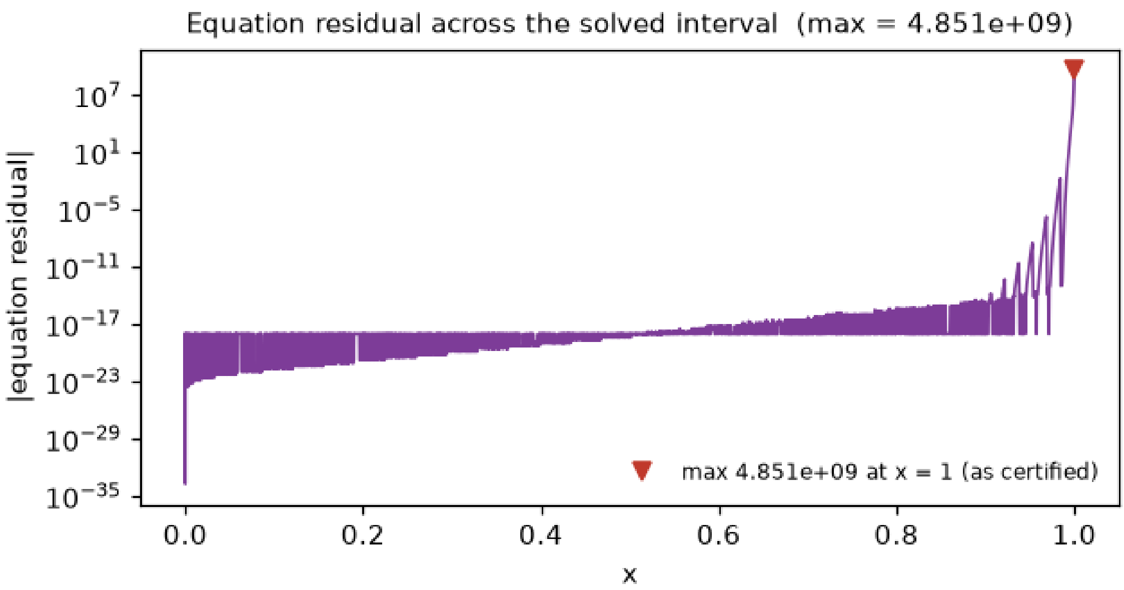
Notes

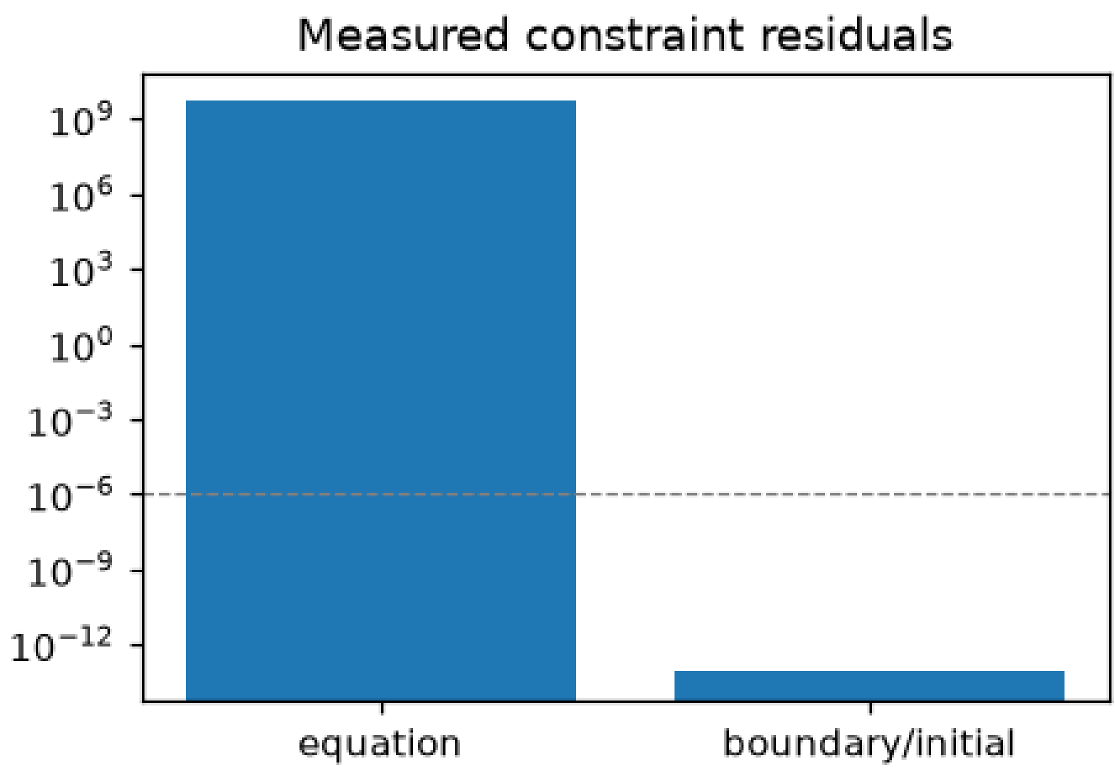
How to read the tail decay ratio. It divides a piece's tail coefficients by its leading ones. Where the solution is very small, the leading coefficients are near zero, so the ratio grows large even though the piece itself is accurate - a large ratio means we divided by something tiny, NOT that the error is large. The authoritative accuracy figures are the measured residuals in Claim 1; if a piece is genuinely stretched, the certificate says so and names the lever (partition / iterations).

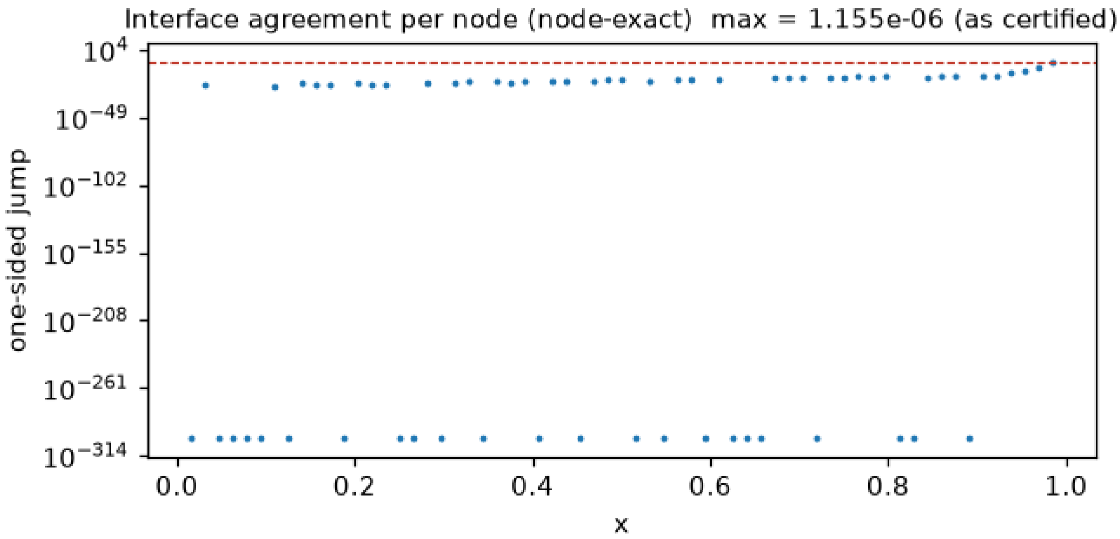
Scope: this tier certifies the four claims above only. Diagnostics, localization, and analytics are Tier-3 deliverables.

Delivered figures




Equation residual across the solved interval (max = 4.851e+09)
|equation residual|
x
max 4.851e+09 at x = 1 (as certified)


Measured constraint residuals
equation
boundary/initial



Issued by LQI. The order manifest (ORDER_AND_DELIVERY.txt) is included in this delivery package alongside this document and the artifact; the terms of service applicable to this order are those supplied with your order confirmation and named by version in the delivery manifest.